

Ballarat High School



Data Analysis – Application Task (2022)

Part 1: Matilda's Bakery



This Data Analysis Application Task is designed to run in 2 parts:

Part 1: Matilda's Bakery (100 Minutes)

Part 2: The Competition (100 minutes)

Introduction

This SAC is looking at analysing data relating to Matilda's journey through the baking industry. Your task is to analyse trends and associations in customer preferences, sales and bakery popularity.

ROUNDING

Unless stated otherwise, round all answers to 1 decimal place

Outcome 1

Define and explain key concepts and apply related mathematical techniques and models as specified in Area of Study 1 in routine contexts. (10 marks)

Outcome 2

Select and apply the mathematical concepts, models and techniques as specified in Area of Study 1 in a range of contexts of increasing complexity. (20 marks)

Outcome 3

Select and appropriately use numerical, graphical, symbolic and statistical functionalities of technology to develop mathematical ideas, produce results and carry out analysis in situations requiring problem-solving, modelling or investigative techniques or approaches. (10 marks)

Section A

As a teen Matilda sold her baked goods on a Sunday afternoon. During this time Matilda collected data about her customers. Using Data Table 1, evaluate the following:

- a) List all the variables from Data Table 1 that are numerical variables. (1)
- b) Create a **percentage two-way frequency table** using the variables *gender* and *slice preference*. (3)
- c) Of the males surveyed, what percentage **did not** prefer vanilla slice? (1)
- d) Use your percentage two-way frequency table to create an appropriate graphical display representing *gender vs slice preferences*. (4)
- e) Is there an association between gender and slice preference? Explain using appropriate evidence. (2)

Section B

After much success as a teen, Matilda opened a bakery. Explore the operation of Matilda's bakery, using Data Table 2.

- a) Find the individual five-number summaries of monthly sales for **each** year 2017, 2018 and 2019. (3)
- b) There is only one outlier in the three years of data. **Show** that the sales for Feb 2018 is an outlier. (2)
- c) Create an appropriate graphical representation that allows the sales figures for 2017, 2018 and 2019 to be compared. (4)
- d) Describe the shape of each distribution. (3)
- e) Which year did Matilda's bakery perform the best? Explain with appropriate evidence. (2)

Section C

Matilda's vanilla slice became famous Ballarat wide.

- a) Matilda's vanilla slice was found to have a mean weight of 173g and a standard deviation of 6.5g. Create an appropriate graphical representation of vanilla slice weight distribution (2)
- b) Shade the graph created to show that 84% of vanilla slices weigh 166.5g or higher? (1)
- c) A customer finds their purchased vanilla slice has a weight of 156g. What is the z-score of this slice? What does this z-score represent in this context? (2)
- d) To ensure a slice with a z-score of **3 or more**, what is the minimum weight (grams) a vanilla slice would need to be? (1)
- e) If Matilda baked 475 vanilla slices, **how many** would be expected to weigh more than 186g? (1)
- f) Matilda decides to reduce the weight of each vanilla slice by 13g. What is the new mean (grams) and standard deviation (grams)? (1)
- g) The actual distribution of vanilla slice weight was found to be negatively skewed. Explain why a normal distribution curve would **not** be reliable. And, given the shape, what can be concluded about the median value of vanilla slice weight? (2)

Section D

Matilda's popularity continues to grow. Data table 3 shows the number of Instagram followers of a selection of bakeries in Victoria.

- a) Tivoli road bakery in Melbourne has been left off the histogram *log(Instagram followers)*. Determine the log number **and** add this bakery to the histogram. (Increase an existing column or add a column as needed) (2)
- b) Determine the median interval of actual Instagram followers for the bakeries. (1)
- c) What percentage of bakeries have less than 1 000 followers? (1)
- d) Which bakery has a log number of Instagram followers of 3.34? (1)